

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI,
K K BIRLA GOA CAMPUS**

Open Book, No electronic devices allowed

Subject Name: BITS F464 - Machine Learning,
Examiner Name: Aditya Challa
Duration: 1.5 hours (2:00 PM - 3:30 PM)

Date: 14 March 2024
Maximum Marks: 48

Instructions

- **Attempt all questions.**
- **Marks corresponding to each question is highlighted in bold within square braces at the end of the question.**
- **You are allowed to carry and refer to any printed material.**
- **In case of ambiguities in any of the questions, clearly state your assumptions and attempt the question(s).**
- **You should write the final answers in the space provided on the question paper. Answers not written in the space provided will not be evaluated.**
- **You can use your answer sheet for rough work.**
- **Return both the question paper and the answer-sheet at the end of the examination.**
- **The exam is for 48 marks. This will be scaled to 30 marks.**

Please fill the following details:

Name:

BITS ID:

BITS Email-ID: (Used for google classroom)

Problem 1

1. In practice, there are a myriad of options as an ML engineer to choose from. While, you cannot readily know if a given option would improve your model, you can usually say which options DO NOT improve your model. Consider the following list of options:
- (A) Increase the regularization coefficient, there by reducing the chance of overfitting.
 - (B) Decrease the regularization coefficient, there by reducing the chance of underfitting.
 - (C) Increase the complexity of the hypothesis class, there by reducing the chance of underfitting.
 - (D) Decrease the complexity of the hypothesis class, there by reducing the chance of overfitting.
 - (E) Issue is with optimization, and you should try a different optimizer/initial conditions.
 - (F) Modify train/test split.
 - (G) Collect more training data and retrain the model.

Assume the following for simplicity:

- (i) You are performing a binary classification task, where the error is the percentage of misclassifications.
- (ii) You can trust the train and test error to be good estimates, i.e they are obtained after averaging over multiple runs.
- (iii) The bias-variance tradeoff principle holds.

- (iv) Ignore the cross-validation paradigm, and assume that test-error is a good estimate of generalization error.
- (v) Also assume that you are doing one thing at a time, and not a combination of things. For instance, if you change train/test split and also change the hypothesis class you might get a different result. This should not be considered.

In each of the following scenarios state which of the options above ((A)-(G)) DO NOT IMPROVE YOUR MODEL.

If you are using any other assumptions apart from above state them here:

- (1a) You find that the train-error is 0 and test-error is 1.

Answer Here:

- (1b) You find that the train-error is 1 and test-error is 0.

Answer Here:

- (1c) You find that the train-error is 0 and test-error for class 1 is 0 and test-error for class 0 is 1.

Answer Here:

- (1d) You find that the train-error is 0.5 and test-error is 0.5.

Answer Here:

- (1e) You find that the train-error is 0 and test-error is 0.

Answer Here:

[3 × 5 = 15 Marks]

Problem 2

2. Let $X = (X_0, X_1, \dots, X_p)$ denote your independent variables where each X_i denotes a feature. A model is called an *additive model* if it can be written as:

$$f(x) = \sum_{i=0}^p f_i(X_i) \quad (1)$$

where each of f_i belongs to some common hypothesis class $\mathcal{H}$. In words, the prediction is the sum of the predictions of the individual features.

State TRUE/FALSE, with justification (in 1-2 lines) for each of the following:

- (2a) Boosting decision stumps (1-level decision trees) is an example of an additive model.

Answer Here:

Justification Here:

- (2b) Boosting decision trees (of depth $d \geq 2$) is an example of an additive model.

Answer Here:

Justification Here:

- (2c) Linear models are an example of an additive model.

Answer Here:

Justification Here:

- (2d) Kernel SVM with kernel $K(x_i, x_j) = x_i^T x_j$ is an example of an additive model.

Answer Here:

Justification Here:

- (2e) Kernel SVM with kernel $K(x_i, x_j) = \exp((\|x_i - x_j\|)/(\sigma^2))$ is an example of an additive model.

Answer Here:

Justification Here:

- (2f) Kernel SVM with kernel $K(x_i, x_j) = \sum_{k=0}^p \gamma_k x_{i,k}^2 x_{j,k}^2$ is an example of an additive model.

Answer Here:

Justification Here:

(2g) Neural networks with ReLU activation functions is an example of an additive model.

Answer Here:

Justification Here:

(2h) Neural networks with Identity activation functions is an example of an additive model.

Answer Here:

Justification Here:

[8 × 2 = 16 Marks]

Problem 3

3. Invariance is usually a strong requirement than is usually given the credit for. Consider the following:

(3a) Let $\{x_0, x_1, x_2, x_3\}$ denote a set and let $\{y_0, y_1, y_2, y_3\}$ denote another set. It is known that a set is invariant to permutations of its elements, i.e the set $\{x_0, x_1, x_2, x_3\}$ is the same as the set $\{x_1, x_0, x_2, x_3\}$. Assume for simplicity that each $x_i, y_i \in \mathbb{R}$.

Can we have a linear classifier that is invariant to permutations of the elements of the set?

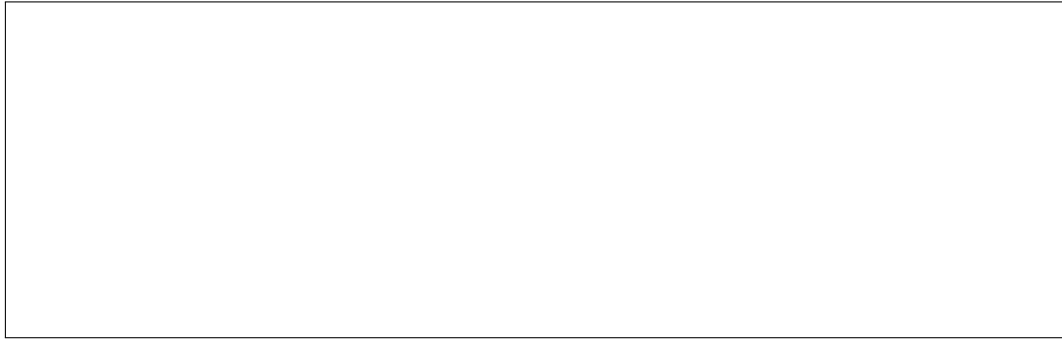
- State TRUE/FALSE and justify your answer - If

$$\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 = \theta_0 x_{\pi(0)} + \theta_1 x_{\pi(1)} + \theta_2 x_{\pi(2)} + \theta_3 x_{\pi(3)} \quad (2)$$

for all permutations π of $\{0, 1, 2, 3\}$ and arbitrary x_0, x_1, x_2, x_3 , then $\theta_0 = \theta_1 = \theta_2 = \theta_3$?

Answer Here:

Justification Here:



- (3b) Where would invariance to permutations be useful? - Recall that a graph is denoted by its vertices and edges - $G_i = (V_i, E_i)$, where $E_i \subset V_i \times V_i$ which is *symmetric* - i.e $(u, v) \in E_i$ if and only if $(v, u) \in E_i$. This is a simple undirected graph.

Two graphs are said to be identical if $V_i = V_j$ and $E_i = E_j$.

Two graphs are said to be isomorphic if there exists a bijection $\phi : V_i \rightarrow V_j$ such that $(u, v) \in E_i$ if and only if $(\phi(u), \phi(v)) \in E_j$.

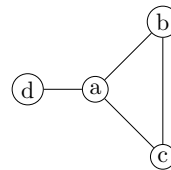
- State TRUE/FALSE - If $G = (V, E)$ is a graph, and π is a permutation of the vertices, then G and G_π are isomorphic.

Answer Here:

- (3c) Consider the following scenario where an oracle is generating random undirected graphs of size n - i.e it randomly connects any two vertices with a probability p . Your job is to check if the generated graph is isomorphic to a given graph G^* .

Assume for simplicity that

- n (number of vertices) is equal to 4
- we want to build a classifier which assigns class 0 if the graph is not isomorphic to G^* and class 1 if the graph is isomorphic to G^* .
- The graph G^* is



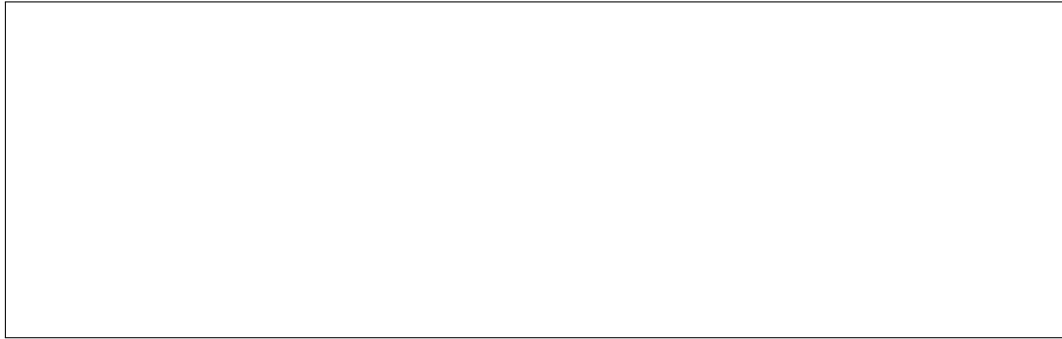
- State TRUE/FALSE - Do the degrees of the vertices identify (as a set) the graph G^* uniquely upto isomorphism? (see figure below for the degrees)

• Answer Here:

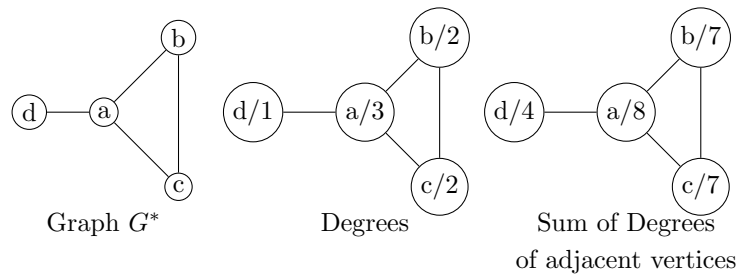
- State TRUE/FALSE with justification - Can we have a linear classifier to solve the problem of checking if the generated graph is isomorphic to G^* using the degrees of the vertices?

Answer Here:

Justification Here:



- (3d) Consider the same scenario as (3c), but instead consider *sum of degrees of adjacent vertices (including itself)* instead of degrees as illustrated in the figure below:



- State TRUE/FALSE - Do the sum of degrees of adjacent vertices identify (as a set) the graph G^* uniquely upto isomorphism?

Answer Here:

- State TRUE/FALSE with justification - Can we have a linear classifier to solve the problem of checking if the generated graph is isomorphic to G^* using the sum of degrees of adjacent vertices?

Answer Here:

Justification Here:

[(1+4)+1 + (2+1+3) + (2+1+3) = 17 Marks]